# Decision report: DEC-20261022-a0f165

GOAL-MONO-001 · lane BATCH-e6e94f (closes BATCH-dce635) · composes REVIEW-MONO-20261017-8aff64 · produced by TASK-20261022-62a47e · 2026-10-22

## Status

This decision is **conditional**. It takes effect only when the ledger archive TASK-20261022-f42d75 commits and verifies with conditions CC-1 to CC-13 holding. Only then does GOAL-MONO-001 move to **completed, on criterion (c) only**. Until then nothing here is official. This is not a performance measurement: no benchmarks/ entry and no index-calculus leaderboard regeneration apply.

## Decision

Decision label, verbatim (DEC-20261022-a0f165 decision_label):

> CONDITIONAL. (1) Route (a) is DECLINED ON THE MERITS: RA-4 composes C, so the J4 DISSENT of TASK-20260930-40470a is sustained on each of its grounds G-i, G-ii and G-iii, and is DISPOSED BY CONCURRENCE under RA-C3 (upheld, not overruled). (2) RA-1, RA-2 and RA-3 compose C and the blind re-derivation RB agrees on Q1 to Q4, so under RA-C5 the DEC-20260930-8553d4 CU-9 derivations, with two mandatory scope corrections, are promoted as the derivation-tier finding KN-FIND-bdcd60, and KN-OPEN-62c870 supersedes KN-OPEN-8a7432. (3) CORR-20261022-87c9cd corrects EV-MONO-e9dc3b's off-vocabulary strength and direction. (4) With criterion (c) recorded met and no DISSENT outstanding, GOAL-MONO-001 moves to COMPLETED ON CRITERION (c) ONLY. All four take effect together, and only if CC-1 to CC-13 hold in the ledger archive TASK-20261022-f42d75.

- **Route (a):** declined on the merits; the J4 DISSENT of TASK-20260930-40470a is sustained and disposed by concurrence.
- **Promotion:** the CU-9 derivations (DEC-20260930-8553d4), with two scope corrections, become KN-FIND-bdcd60; KN-OPEN-62c870 supersedes KN-OPEN-8a7432.
- **Correction:** CORR-20261022-87c9cd corrects EV-MONO-e9dc3b (H-MONO-ebd400; EXP-MONO-0e6e8f, RUN-MONO-0e6e8f-1).
- **Evidence strength:** replicated, by CORR-20261022-87c9cd for EV-MONO-e9dc3b; derivation tier for KN-FIND-bdcd60. **Claim tier:** toy.
- **Scoped claim (supports), verbatim:** The CU-9 derivations as corrected, scoped as KN-FIND-bdcd60 scope_frozen: p > 3, nonsingular E/F_p, generic monodromy, every m >= 3 by derivation (symmetric-base order formula m >= 4), measured only at m <= 5 and the listed toy primes.
- Records: DEC-20261022-a0f165, DEC-20261017-435b01, DEC-20261017-12e0f4, EV-MONO-e9dc3b, H-MONO-ebd400, EXP-MONO-0e6e8f, RUN-MONO-0e6e8f-1, TASK-20261017-ccd10c, TASK-20261017-8aff64, TASK-20261017-fb4481, TASK-20260930-40470a.

## Observation

The review round REVIEW-MONO-20261017-8aff64 ran two fresh max-effort sessions with zero runs: the blind re-derivation TASK-20261017-ccd10c (validator-breakthrough, joint RB) and the joint review TASK-20261017-8aff64 (red-team-breakthrough, joints RA-1 to RA-4). Both outputs were snapshot-archived by TASK-20261017-fb4481 (commit e3de77970e64). Every plotted value below is read by key from their computations.json and checked against the report table it comes from (plot-data.json, 75 cells). The two reports reuse the labels C1 to C8 for different curves; each row names its source and (p, a, b). Wilson 95% intervals are derived here from the counts.

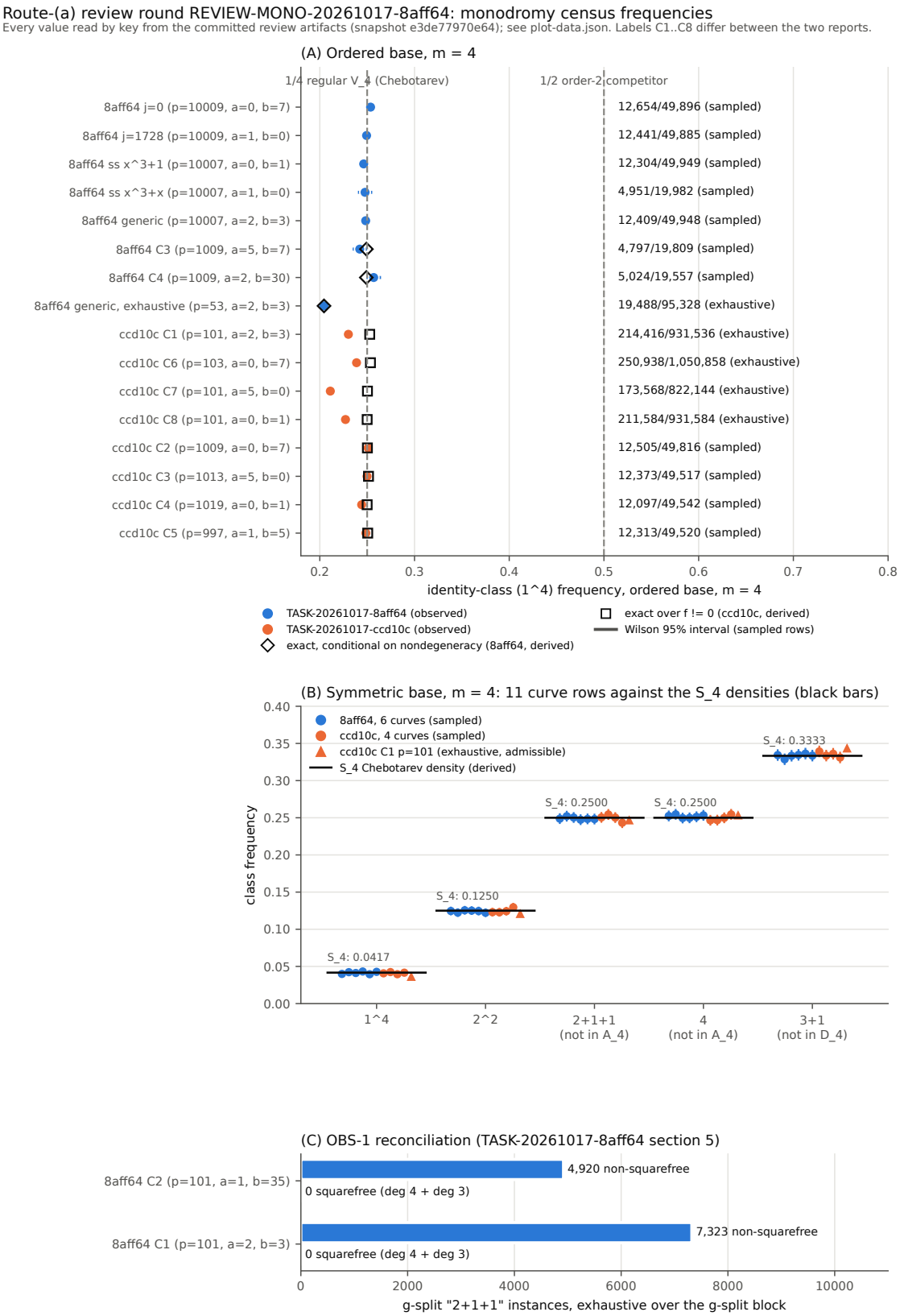



Figure 1: Panel A: ordered base, identity-class frequency at m = 4 against 1/4 (regular V_4) and 1/2 (order-2 competitor), with the exact finite-p expectations each report gives (diamond: exact conditional on nondegeneracy, 8aff64; square: exact over f ≠ 0, ccd10c). Panel B: symmetric base, the five class frequencies against the S_4 densities, classes mapped by label. Panel C: the OBS-1 g-split "2+1+1" instances on C1 and C2, all non-squarefree. Data: plot-data.json; regenerate with plot.py.

Not plotted (report table only; not traced by key): 8aff64 section 2.4 second table: the 'exact conditional' values for p near 10^4 are given only as approximate bounds (e.g. '≈ 0.2499'); they are not plotted as expectations. 8aff64 section 2.4: the 200k-tuple p=1009 samples (big1009_out.json) are reported in the second table, not the first; panel A plots the first table's rows as the card specifies..

## Comparison

Joint table (DEC-20261022-a0f165 composition.joint_table):

| joint | owner | verdict as reported | composed status |
|---|---|---|---|
| RA-1 ordered-base monodromy (CU-9 item 1) | TASK-20261017-8aff64 | CONCUR (holds) | C |
| RA-2 symmetric-base monodromy (CU-9 item 2) | TASK-20261017-8aff64 | CONCUR (holds) | C, with the corrected wording mandatory in every adopting text (CS-PR, CS-M3) |
| RA-3 the delta-rule (CU-9 item 3) | TASK-20261017-8aff64 | CONCUR (holds) | C |
| RA-4 J4 adequacy | TASK-20261017-8aff64 | CONCUR (holds) | C |
| RB blind determination | TASK-20261017-ccd10c | CONCUR (holds) | C (independent; RA-C6 not triggered; ruling I-2) |

K4b comparison, RA (8aff64) against RB (ccd10c), quantity by quantity:

| quantity | RA | RB | agreement |
|---|---|---|---|
| Q1 ordered base | (Z/2)^{m-2}, order 2^{m-2}, regular, one orbit; arithmetic = geometric | C_2^n / <-1> = (Z/2)^{m-2}, regular, elementary abelian; arithmetic = geometric | full |
| Q2 symmetric base | (Z/2)^{m-2} semidirect S_{m-1}, kernel {+-1}, order 2^{m-2}(m-1)! for m >= 4; S_4 at m = 4; 192, 1920, 23040 at m = 5, 6, 7; primitive at m = 4, 6; one block system of two blocks at m = 5, 7; S_2 at m = 3 | orders 2, 24, 192, 1920, 23040 at m = 3..7 (kernel of order 4 at m = 3); S_4 at m = 4; primitive exactly at m = 3, 4, 6; unique nontrivial block system, two blocks (parity classes), at m = 5, 7 | full, including both corrections |
| Q3 Frobenius rule | 1^{2^{m-2}} iff delta = +-(1..1), otherwise a fixed-point-free involution | 1^{2^{m-2}} iff all chi(f(x_i)) are equal, otherwise 2^{2^{m-3}} | full (the same statement) |
| Q4 curve dependence | none; j = 0, j = 1728 and supersingular curves tested | none; a = 0, b = 0 and supersingular curves tested; only finite-p counts depend on the curve | full |

Result: Agreement on every determination. Nothing to localise. K4b has no effect beyond making the two corrections mandatory wording.

## Inference

As decided in DEC-20261022-a0f165, not added here. Goal-status criteria (goal_status_ruling.criteria):

| | |
|---|---|
| **(a)** | NOT MET. Declined on the merits (G-i, G-ii, G-iii sustained). The route-(a) object satisfies only the exact-identification reading R-x on the symmetric base, not "within stated error" and not KN-OPEN-009's ordered base. |
| **(b)** | NOT MET. No exceptional locus is exhibited. For generic monodromy none exists for any nonsingular curve (KN-FIND-bdcd60); nothing is said about rates. |
| **(c)** | MET, as recorded by DEC-20261017-435b01, in effect since the verified ledger archive 5a11b9500982: the scoped instrument no-go KN-FIND-4f0d84, read with CS-D2. This decision does not re-adopt it and adds nothing to its scope. |

Goal status on effect: completed (on effect) conditional on CC-1 to CC-13 in TASK-20261022-f42d75.

## Independence and disclosure rulings

- **I-1** (SA-9 check_review_independence.py exit 1, one problem): A SCHEMA DEFECT, NOT AN INDEPENDENCE FAILURE, conditional on CC-4. The seven entries at indices 12, 13, 17, 39, 47, 48 and 49 of the 8aff64 sources_read each contain an unquoted ': ' and parse as one-key mappings. This composer read each entry; they name: EXP-MONO-0e6e8f raw-result.json (by key), EXP-MONO-815525 run_census.py (lines 47-209), TASK-20260930-40470a s4_construction/computations.json (by key), templates/research-records.md (lines 675-789), agents/coordinator.md (lines 178-247), docs/claims-and-verification.md (lines 104-170) and EXP-MONO-815525 red-team-report.yaml (lines 256-300). None is a sibling artifact, the prior-holding decision or an excluded path. The checker also SKIPS non-string entries in its sibling test (tools/check_review_independence.py lines 953-966, `continue` on non-str), so its clean pass on the other entries does not cover these seven. CC-4 closes that gap mechanically on a normalized scratch copy. The committed file is not edited (SA-7; records immutable).
- **I-2** (TASK-20261017-ccd10c read .claude/agents/validator.md and docs/inventor-protocol.md, outside its whitelist): A DISCLOSED READ-SCOPE DEVIATION (RB-9), CAUSED BY THE CARD, WITH NO EFFECT ON BLINDNESS. Neither path is under any blind_from entry (the list names coordination/, ledger/, knowledge/, experiments/, analysis/ and research/ paths; docs/ and .claude/ are not in it), so RA-C6's not-independent clause is not triggered and the checker correctly raised nothing for ccd10c. Neither file contains the quantities: this composer's search of each for monodromy, Semaev, V_4 and summation found no match (and S_4 none in validator.md). The binding text the re-deriver had to read (.claude/agents/validator-breakthrough.md lines 27-32) directs the reader to both files, and the card's whitelist omitted them: a card-design defect of the opening Coordinator session, not a reviewer defect. Lesson LS-4.
- **I-3** (TASK-20261017-8aff64 read four decision ranges outside its inputs for Q-OI): A DISCLOSED READ-SCOPE EXCESS, PURPOSE-BOUND, WITH NO EFFECT ON COMPOSITION. DEC-20260930-3a0b7b lines 184-194, DEC-20260904-0f525b lines 40-66, DEC-20260904-4eb323 lines 184-192 and DEC-20260904-8c2580 lines 118-156 are grep-located single ranges, declared by path and

line. None is an excluded path (the excluded set is DEC-20261017-12e0f4, the ccd10c directory and records, EXP-MONO-4b50b6/ and BATCH-003/), none is a sibling artifact, and none holds the coordinator prior. This composer read DEC-20260904-8c2580 lines 118-156, DEC-20260802-a51c82 N-3 and DEC-20260810-2f86db N-3 (which carries the a51c82 N-3 repairs forward and adds "No new census of any statistic may be dispatched before these are carried") and finds the reviewer's quotations accurate. The grep -l listing that printed three file NAMES under the excluded BATCH-003/ path, with no content, is name-level exposure only; 8aff64 is not a blind re-deriver and BATCH-003 is kept closed for later blind rounds, of which this goal now has none. Lesson LS-6.

- **I-4** (the mutual-blindness measure (holding folder) and the shared scratchpad): ACCEPTED AS A CUSTODY MEASURE OUTSIDE THE PLAN (PD-1); MUTUAL BLINDNESS OF 8aff64 FROM RB RESTS ON ATTESTATION, CORROBORATED; RB's BLINDNESS FROM 8aff64 IS STRUCTURAL. (a) Custody: the moved files were restored with identical sha256 (runtime receipt) and equal the release digests (SA-4), so no byte changed. (b) Direction RB -> RA: ccd10c ran and was released before 8aff64 was claimed, so it could not see 8aff64's output. (c) Direction RA -> RB: the measure kept the sibling's output out of the working tree, and 8aff64 reports that ccd10c's directory did not exist in its checkout. But the ccd10c runtime receipt also discloses that all its scripts were written and run in the shared session scratchpad, and 8aff64 attests that it listed foreign scratchpad directories (blind/, rb/, held/, ...) by name once and never opened them. That residual exposure cannot be checked after the fact; it rests on 8aff64's attestation. It is corroborated by method divergence: different curves and primes (8aff64 p = 53, 59, 101, 1009, 10007, 10009; ccd10c p = 101, 103, 997, 1009, 1013, 1019, sharing only the (101, 2, 3) curve), different S_5 constructions (per-fibre 12-point interpolation of Res_X(S_4, S_3) against full-grid interpolation of the formal Sylvester determinant), different exact-count methods (enumeration of the degenerate locus against closed-form class counts), and different S_4 term counts (439 monomials in x with Z[a,b] coefficients against 540 terms; a counting-convention difference, not reconciled here, not a Q1-Q4 quantity and relied on by neither). Composition consequence: RB is independent of both the producer and 8aff64; RA-1 to RA-3 count as a second re-derivation independent of RB by attestation and divergence, not by construction. Lesson LS-3.
- **I-5** (models reported as claude-fable-5-1, not probe-verified): RECORDED AS REPORTED, NEVER AS VERIFIED. Both review sessions report claude-fable-5-1, the binding's pin; this composer and the producers report claude-opus-5-5. model_verified is false everywhere; no doctor --probe was run. Independence is by session and role, and between the two reviewers by task; by model only as reported, and the two reviewers share one model. This caps the strength this decision may assign at replicated, never strong (CS-IND).
- **I-6** (ccd10c's first-run log overwritten by a re-run (blind_rederivation.md section 7)): DISCLOSED, NOT RELIED ON. The embedded logs and outputs are from the deterministic re-run; the first run's output exists only in the session transcript and is not evidence. No composed value depends on it.
- **I-7** (the coordinator prior): Isolated as designed (LS-1). Neither attestation lists DEC-20261017-12e0f4; for ccd10c it is a blind_from path that the checker tests and found unread. CC-5 checks mechanically that the marker PR8- appears in none of the six review files.

Result: The round is independent as the plan designed it: the blind re-derivation is a separate task, blind to every blind_from path and structurally blind to the joint owner; the joint owner is blind to the re-derivation by attestation, corroborated. No reviewer read a sibling report. The checker's one problem is a schema defect whose blind spot CC-4 closes.

## Limitation

- **L-1**: CS-IND. One model family served both reviewers; no identifier is probe-verified; 8aff64's blindness from RB rests on its attestation and on method divergence.
- **L-2**: One reviewer owned RA-1 to RA-4, the control and both questions. The cross-reviewer independence this round adds is on the load-bearing quantities (RB), not on RA-4, which has a single owner.
- **L-3**: Toy primes only (53 to 10009). Exact finite-p expectations were enumerated only at p <= 1009 (8aff64) and by closed form (ccd10c); two of about thirty goodness-of-fit p-values in ccd10c are below 0.05 and 8aff64 reports single-sample excursions of 2.3 to 2.7 sigma, each within its declared bound and explained in its report.
- **L-4**: This composer holds no shell. Digests, checker output and git facts are taken from committed receipts and re-checked by CC-1 to CC-13. It read computations.json only by one key search and did not re-run any review code.
- **L-5**: Semaev 2004 is a recalled pointer in both reports; the two facts used (defining property, resultant recursion) were verified numerically at m = 3, 4, 5 by each reviewer. The m >= 6 extent rests on the derivations and a curve-free group computation.
- **L-6**: PF-1 is recorded in this decision, not in a separate correction record; the round's one reserved correction id went to EV-MONO-e9dc3b.
- **L-7**: Literature dominance of the group determinations was not checked (honest_accounting.dominated_by).

## Counter-statements (carried verbatim)

**CS-K1d2**

> Nothing in this decision or its projections states or implies that GOAL-MONO-001 criterion (a) is met. On KN-OPEN-009's ordered base the generic monodromy is (Z/2)^{m-2} acting regularly, for every nonsingular curve; it is not full symmetric or wreath. The completion rests on criterion (c) alone.

**CS-PR**

> H-MONO-93bc4d (statement and claim_boundary "a proper imprimitive subgroup of Sym(2^{m-2}) at m>=5"), DEC-20260904-dd58c3's limitation ("PROPER, imprimitive") and DEC-20260930-8553d4 CU-9 ("primitivity varying for m = 5..7") are read with the corrected wording: for m >= 4 the symmetric-base group (Z/2)^{m-2} semidirect S_{m-1} is primitive iff m - 1 is odd. It is primitive at m = 4 (S_4) and at m = 6 (order 1920, proper); at m = 5 and m = 7 it is imprimitive with exactly one nontrivial block system, two blocks of size 2^{m-3} (the parity classes of the number of minus signs).

**CS-M3**

The order 2^{m-2}(m-1)! of the symmetric-base group holds for m >= 4. At m = 3 the symmetric-base group is S_2 of order 2 (the kernel has order 4), as CORR-20260904-d2c631 records for H-MONO-93bc4d (D3).

**CS-QA2**

CS-QA of DEC-20261017-435b01 says the uniform-in-curve statement is "so far a single-reviewer derivation (DEC-20260930-8553d4 CU-9) that no committed record has re-derived". That was true when written. From this decision's verified archive the CU-9 derivations are re-derived, blind (TASK-20261017-ccd10c) and independently (TASK-20261017-8aff64), and recorded as KN-FIND-bdcd60 at derivation tier. CS-QA's operative content stands: DEC-20260904-dd58c3's census covers 4 curves at 2 primes, and neither it nor the census establishes criterion (a).

**CS-R7**

Toy tier. Nothing here concerns the relation-rate question, the D-4 factor-base-point count beyond "not condemned", extension fields or cryptographic scale. No exponent moves and no security estimate moves.

**CS-IND**

Independence in REVIEW-MONO-20261017-8aff64 is by session and role. The blind re-derivation (TASK-20261017-ccd10c) ran first, read no blind_from path and could not see the joint review. The joint owner (TASK-20261017-8aff64) attests that it never opened the re-deriver's files, which sat in a scratchpad it could list; that rests on its attestation and on method divergence. Both reviewers report claude-fable-5-1 and the composer claude-opus-5-5; no identifier is probe-verified.

## Canonical graphs checked

knowledge/frontiers/ecdlp/: README.md, KR-RHO-53c89f, KR-IC-5931ee, KR-IC-5a43e5, KR-IC-889857 and KR-IC-fbdb61 mention Semaev polynomials (cost bounds, assumptions, decomposition); none mentions monodromy or GOAL-MONO-001. All were checked and are **unchanged**: this decision is a toy-tier derivation finding with no cost claim. No scope finding.

## Next actions

**NA-1** (coordinator (shell-holding top-level session; archive and mechanical production; no status authority)):

Commit the BATCH-e6e94f queue alone and open lane BATCH-e6e94f (f42d75 LA-1). Pin the sha256 of the eleven working-tree files of this act (CC-11). Run TASK-20261022-62a47e (decision report, graph and PDF) and release it with --artifact path=sha256. Then run the ledger archive TASK-20261022-f42d75 alone, checking CC-1 to CC-13. Verify it, push, and open or refresh the PR naming every record id. Then close lanes BATCH-dce635 and BATCH-e6e94f with outcome archived, naming the f42d75 commit.

Declared successor:

None on GOAL-MONO-001: on effect it is completed. Outside this goal, the next portfolio /coordinate ranks IDEA-20260803-ff7415 (re-scope by the idea-generator anchored on KN-OPEN-62c870 Q3, per DEC-20261017-12e0f4 rerank.successor_after_this_lane) and the KN-OPEN-62c870 Q1 leads under "ECC comes first".